ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

SMOOTHING IN N-GRAM LANGUAGE MODELS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

Smoothing N-Gram Language Models

Please close the first door on the left.

4-Gram	3-Gram	2-Gram
3380 please close the door	197302 close the window	198015222 the first
1601 please close the window	191125 close the door	194623024 the same
1164 please close the new	152500 close the gap	168504105 the following
1159 please close the gate	116451 close the thread	158562063 the world
…	…	…
0 please close the first	8662 close the first	…
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13951 please close the *	3785230 close the *	23135851162 the *
0.0	0.002	0.009

Specific but Sparse ⟷ Dense but General

Simplest way to mix different orders: linear interpolation

$$\lambda \hat{P}(w|w_{-1}, w_{-2}) + \lambda' \hat{P}(w|w_{-1}) + \lambda'' \hat{P}(w)$$

- How to choose lambdas?
- Should lambda depend on the counts of the histories?

Choosing weights: either grid search or EM using held-out data

Better methods have interpolation weights connected to context counts, so you smooth more when you know less

Observation: N-grams occur more in training data than they will later

Empirical Bigram Counts (Church and Gale, 91)

Count in 22M Words	Future c* (Next 22M)
1	0.45
2	1.25
3	2.24
4	3.23
5	4.21

Absolute discounting

- Reduce numerator counts by a constant d (e.g. 0.75)
- Maybe have a special discount for small counts
- Redistribute the “shaved” mass to a model of new events

Example formulation

$$P_{\text{ad}}(w|w') = \frac{c(w', w) - d}{c(w')} + \alpha(w')\hat{P}(w)$$

Shannon game: “There was an unexpected _____”

- “delay”?
- “Francisco”?

Context fertility: number of distinct context types that a word occurs in

- What is the fertility of “delay”?
- What is the fertility of “Francisco”?
- Which is more likely in an arbitrary new context?

Kneser-Ney smoothing combines two ideas

- Discount and reallocate like absolute discounting
- In the backoff model, word probabilities are proportional to context fertility, not frequency

$$P(w) \propto |\{w' : c(w', w) > 0\}|$$

Theory and practice

- Practice: KN smoothing has been repeatedly proven both effective and efficient
- Theory: KN smoothing as approximate inference in a hierarchical Pitman-Yor process [Teh, 2006]

WHAT ACTUALLY WORKS?

Trigrams and beyond:

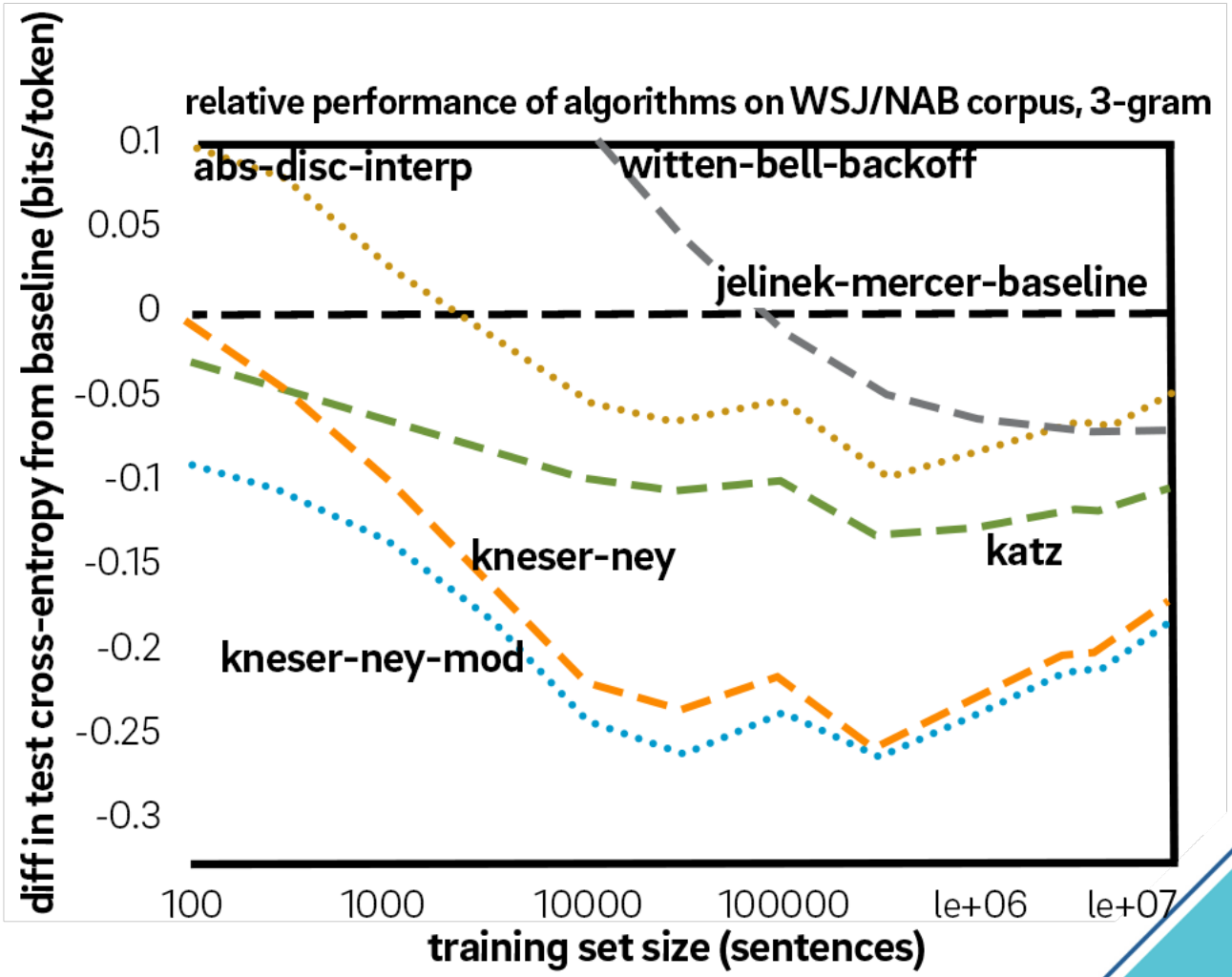
- Unigrams, bigrams generally useless
- Trigrams much better
- 4-, 5-grams and higher are good enough to be practically useful

Discounting

- Absolute discounting, Good-Turing, held-out estimation, Witten-Bell, etc…

Context counting

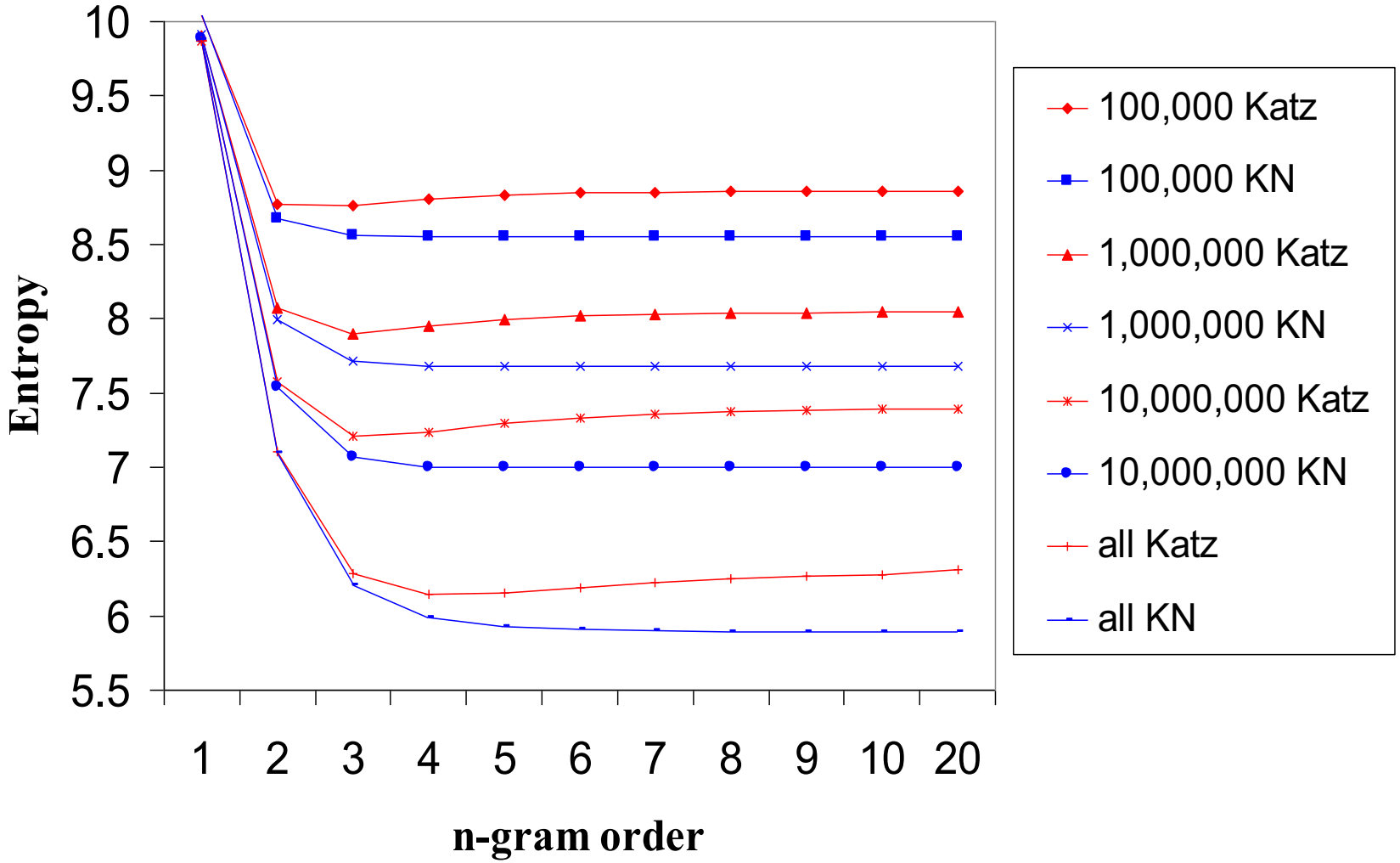
- Kneser-Ney construction of lower-order models



[Graph from Joshua Goodman]

N-Gram Models from Big Data

Having more data is better…



… but so is using a better estimator

Another issue: N > 3 has huge costs in some speech recognizers

Just about every local correlation!

- Word class restrictions: "will have been ___"
- Morphology: "she ___", "they ___"
- Semantic class restrictions: "danced the ___"
- Idioms: "add insult to ___"
- World knowledge: "ice caps have ___"
- Pop culture: "the empire strikes ___"

But not the long-distance ones

- "The computer which I had just put into the machine room on the fifth floor ___."